HAONAN CHEN

Kempner Institute ⋄ 150 Western Avenue Allston, MA 02134, U.S.
haonan_chen@seas.harvard.edu https://haonan16.github.io/

EXPERIENCES

Postdoctoral Research Fellow
Harvard University 09/2025 – Present
Advisor: Yilun Du

EDUCATION

Ph.D. in Electrical and Computer Engineering
University of Illinois Urbana-Champaign 08/2020 - 08/2025

B.Sc. in Electrical Engineering with Highest Honors
University of Illinois Urbana-Champaign 09/2016 - 05/2020
GPA: 3.98/4.0

B.Eng. in Electrical Engineering and Automation
Zhejiang University 09/2016 - 06/2020
GPA: 3.99/4.0 (Rank: 1/30)

JOURNAL PUBLICATIONS

[**RA-L'26**] C. Liu, H. Chen, S. H. Høeg*, S. Yao*, Y. Li, K. Hauser, Y. Du. "Flexible Multitask Learning with Factorized Diffusion Policy." *IEEE Robotics and Automation Letters,* 2026. [pdf] [Website]

[**ITSM'24**] A. Hasan, N. Chakraborty, H. Chen, J.-H. Cho, C. Wu, K. Driggs-Campbell. "Lessons in Cooperation: A Qualitative Analysis of Driver Sentiments towards real-time Advisory Systems through a Focus Group User Study." *IEEE Intelligent Transportation Systems Magazine,* 2024. [pdf]

[**JATS'24**] A. Hasan, N. Chakraborty, H. Chen, J.-H. Cho, C. Wu, K. Driggs-Campbell. "Cooperative Advisory Residual Policies for Congestion Mitigation." *ACM Journal on Autonomous Transportation Systems*, 2024. [pdf]

PREPRINTS

C. Gao, W. Huang, H. Chen, L. Shao, Y. Du, J. Wu, R. Zhang, L. Fei-Fei. "Dream What I Can: Sparse Visual Planning for Long-Horizon Robot Manipulation." *arXiv preprint,* 2026. [pdf] [Website] [Code]

Y. Ma, W. Xu, P. Ye, A. Mei, Y. R. Song, T. Li, H. Chen, E. H. Adelson. "ACTS: Learning Visuo-Tactile World Models from Human Physical Interaction." *arXiv preprint,* 2026. [pdf] [Website]

H. Chen*, F. Wu*, Y. Ma, M. Mete, P. Ye, J. Shen, C. Zhu, A. Ruggeri, K. Cheung, J. Mao, E. Adelson, J. Wu, R. D. Howe, Y. Du. "VisualForce: Robot Learning with Visual Predicted Force." *arXiv preprint,* 2026. [pdf] [Website] [Code]

R. Sathua, H. Chen, K. Driggs-Campbell. "ARGUS: Aligning Robot Scene Geometry Under Shifting Views with Large 3D Vision Models." *arXiv preprint,* 2026. [pdf] [Website]

S. Lu, H. Chen, W. Ye, Y. Du. "DriftWorld: Fast World Modeling through Drifting." *arXiv preprint,* 2026. [pdf] [Website] [Code]

X. Han*, H. Xiong*, H. Chen, C. Liu, A. Torralba, Y. Zhu, Y. Du. "B-spline Policy: Accelerating Manipulation Policies via B-spline Action Representations." *arXiv preprint,* 2026. [pdf] [Website] [Code]

P. Ye, Y. Ma, H. Chen, G. Wang, Y. Jing, B. Sheil, Y. Du, E. Adelson. "You Only Touch Once: 6-DoF Object Pose Estimation from Single Tactile Contact." *arXiv preprint,* 2026. [pdf]

J. Gao, C. Liu, P. Wu, H. Chen, Y. Du. "IMPACT: Learning Internal-Model Predictive Control for Forceful Robotic Manipulation." *arXiv preprint,* 2026. [pdf] [Website] [Code]

PEER REVIEWED CONFERENCE PUBLICATIONS

[CoRL'26] A. Laprevotte, H. Chen, X. Han, Y. Du. "Learning to Be Different: Component Specialization for Compositional Diffusion Policies." *Conference on Robot Learning,* 2026. [pdf] [Website] [Code]

[Humanoids'26] Y. Ma*, M. Tippur*, P. Ye, S. Q. Liu, H. Chen, F. R. Cottrell, E. Adelson. "GlowTact: Simple and Compact Vision-Based Tactile Sensing with High Sensitivity and Spatial Resolution." *IEEE-RAS International Conference on Humanoid Robots,* 2026. [pdf] [Website]

[NeurIPS'26] H. Alzayer, W. Huang, H. Chen, C. Luey, L. Zhang, M. Agrawala, G. Wetzstein, L. Fei-Fei, Y. Du, J. Wu, J.-B. Huang. "Masked Visual Actions for Unified World Modeling." *Advances in Neural Information Processing Systems,* 2026. [pdf] [Website] [Code]

[CoRL'26] H. Chen*, Y. Ma*, S. Tian, X. Han, W. Huang, F. Wu, Y. Li, J. Wu, E. H. Adelson, Y. Du. "CoStream: Composing Simple Behaviors for Generalizable Complex Manipulation." *Conference on Robot Learning,* 2026. [pdf] [Website]

[ECCV'26] H. Chen, K. Zhou, H. Hua, K. Zhang, J. Qian, W. Ma, H. Chen, C. Liu, Y. Zhao, X. Wang, W. Li, A. Yuille, P. P. Liang, Y. Du. "MemoBench: Benchmarking World Modeling in Dynamically Changing Environments." *European Conference on Computer Vision,* 2026. [pdf] [Website] [Code]

[RSS'26] C. Liu, X. Han, J. Gao, Y. Zhao, H. Chen, Y. Du. "OAT: Ordered Action Tokenization." *Robotics: Science and Systems,* 2026. [pdf] [Website] [Code]
Finalist - Best/Outstanding Paper Awards

[CVPR'26] H. Liu*, S. Yao*, H. Chen, J. Gao, J. Mao, J.-B. Huang, Y. Du. "SIMPACT: Simulation-Enabled Action Planning using Vision-Language Models." *IEEE/CVF Conference on Computer Vision and Pattern Recognition,* 2026. [pdf] [Website]

[ICRA'26] K. Hong, H. Chen*, J. Xu*, R. Wang*, K. Wang, M. Zhang, S. Liu, Y. Zhu, Y. Li, K. Driggs-Campbell. "Gotta Scoop 'Em All: Sim-and-Real Co-Training of Graph-based Neural Dynamics for Long-Horizon Scooping." *IEEE International Conference on Robotics and Automation,* 2026. [pdf] [Website]

[ICRA'26] H. Chen, J. Xu*, H. Chen*, K. Hong, B. Huang, C. Liu, J. Mao, Y. Li, Y. Du+, K. Driggs-Campbell+. "Multi-Modal Manipulation via Multi-Modal Policy Consensus." *IEEE International Conference on Robotics and Automation,* 2026. [pdf] [Website] [Code]
Best Paper Award at [CVPR 2026 Workshop on Multi-Sensory Modeling for Embodied Intelligence]
Featured in Video Friday on [IEEE Spectrum]

[CoRL'25] H. Chen, C. Zhu, S. Liu, Y. Li, K. Driggs-Campbell. "Tool-as-Interface: Learning Robot Policies from Observing Human Tool Use." *Conference on Robot Learning,* 2025. [pdf] [Website] [Code]
Best Paper Award, [ICRA 2025 Workshop on Foundation Models and Neuro-Symbolic AI for Robotics]
Best Presentation Award at [CSL Student Conference 2025]

[RSS'25] S. Peng, H. Chen, K. Driggs-Campbell. "Towards Uncertainty Unification: A Case Study for Preference Learning." *Robotics: Science and Systems,* 2025. [pdf] [Website] [Code]
Best Paper Award at [ICRA 2026 Workshop on Uncertainty in Open-World Robotics]

[ICRA'25] H. Chen, J. Xu*, L. Sheng*, T. Ji, S. Liu, Y. Li, K. Driggs-Campbell. "Learning Coordinated Bimanual Manipulation Policies using State Diffusion and Inverse Dynamics Models." *IEEE International Conference on Robotics and Automation,* 2025. [pdf] [Website]

[CoRL'23] H. Chen, Y. Niu*, K. Hong*, S. Liu, Y. Wang, Y. Li, K. Driggs-Campbell. "Predicting Object Interactions with Behavior Primitives: An Application in Stowing Tasks." *Conference on Robot Learning,* 2023. [pdf] [Website]
Finalist - Best Paper/Best Student Paper Awards

[CASE'23] Z. Huang, Y. Mun, H. Chen, Y. Xie, Y. Niu, X. Li, N. Zhong, H. You, D.L. McPherson, K. Driggs-Campbell. "Towards Safety of Multi-Level Human-Robot Interaction in Industrial Tasks." *IEEE Conference on Automation Science and Engineering (CASE), Special Session on Cyberphysical Manufacturing Networks,* 2023. [pdf]

[CASE'23] H. Chen*, Y. Mun*, Z. Huang, Y. Niu, Y. Xie, D.L. McPherson, K. Driggs-Campbell. "Learning Task Skills and Goals Simultaneously from Physical Interaction." *IEEE Conference on Automation Science and Engineering (CASE), Special Session on Cyberphysical Manufacturing Networks,* 2023. [pdf]

[ITSC'23] A. Hasan, N. Chakraborty*, H. Chen*, J-H. Cho, C. Wu, K. Driggs-Campbell. "PeRP: Personalized Residual Policies For Congestion Mitigation Through Co-operative Advisory Systems." *IEEE International Conference on Intelligent Transportation Systems,* 2023. [pdf] [Website]
Best Presentation Award at [CSL Student Conference 2024]

[ITSC'22] H. Chen, T. Ji, S. Liu, K. Driggs-Campbell. "Combining Model-Based Controllers and Generative Adversarial Imitation Learning for Traffic Simulation." *IEEE International Conference on Intelligent Transportation Systems (ITSC 2022).* [pdf]

[ICRA'22] S. Liu, P. Chang, H. Chen, N. Chakraborty, K. Driggs-Campbell. "Learning to Navigate Intersections with Unsupervised Driver Trait Inference." *IEEE International Conference on Robotics and Automation,* 2022. [pdf] [Website]

[IROS'20] P. Chang, S. Liu, H. Chen, K. Driggs-Campbell. "Robot Sound Interpretation: Combining Sight and Sound in Learning-Based Control." *IEEE/RSJ International Conference on Intelligent Robots and Systems,* 2020. [pdf] [Video]

INVITED TALKS & PANELS

July 2026 Professor Zackory Erickson's Group, Carnegie Mellon University

April 2026 Professor Paul Liang's Group, MIT
Title: *Multi-Modal Action Composition for Generalizable Robot Manipulation*

October 2025 Computational Robotics Lab, Harvard University
Title: *Multi-Modal Action Composition for Generalizable Robot Manipulation*

June 2025 Tacta Systems
Title: *Learning What, How, and Where from Humans: Structured Inductive Biases for Robot Learning*

June 2025 Professor Jiajun Wu's Lab, Stanford University
Title: *Learning What, How, and Where from Humans: Structured Inductive Biases for Robot Learning*

May 2025 Professor Roberto Martín-Martín's Lab, University of Texas at Austin
Title: *Toward Human-Robot Alignment through Human Behavior Priors*

May 2025 St. Matthew Catholic School (Outreach Talk to Grade 5 Students)
Title: *Toward Human-Robot Alignment through Human Behavior Priors*

April 2025 Professor Anima Anandkumar's Lab, Caltech
Title: *Deployable Robot Learning from Human Priors*

April 2025 Professor Katerina Fragkiadaki's Lab, Carnegie Mellon University
Title: *Deployable Robot Learning from Human Priors*

March 2025 Safe AI Lab, Carnegie Mellon University
Title: *Building Deployable and Pragmatic Robots from Human Priors*

March 2025 Parasol Lab, University of Illinois Urbana-Champaign
Title: *Tool-as-Interface: Learning Robot Tool Use from Human Play through Imitation Learning*

MEDIA COVERAGE

- **Multi-Modal Manipulation via Multi-Modal Policy Consensus** — Featured in Video Friday, IEEE Spectrum
- **OAT: Ordered Action Tokenization** — MarkTechPost
- **Tool-as-Interface** — Featured in Video Friday, IEEE Spectrum; The Grainger College of Engineering; TechXplore; Hackster.io

TEACHING EXPERIENCE

Graduate Teaching Assistant

- **UIUC ECE 484:** Safe Autonomy **Spring 2025**
  Instructor: Prof. Sayan Mitra

Undergraduate Teaching Assistant

- **ZJU-UIUC Math 241:** Calculus III **Fall 2018**
  Instructor: Prof. Thomas Honold
- **ZJU-UIUC ECE 310:** Digital Signal Processing
  ZJU-UIUC ECE 311: Digital Signal Processing Lab **Fall 2019**
  Instructor: Prof. Mark D. Butala

MENTORED STUDENTS

Undergraduate Students

- Jiaming Xu — Placement: University of Illinois Urbana-Champaign
- Lily Sheng — Placement: Tsinghua University
- Yilong Niu — Placement: University of Illinois Urbana-Champaign
- Cheng Zhu — Placement: University of Pennsylvania
- Eric Ge — Harvard University
- Celinda Zhu — Massachusetts Institute of Technology
- Ashely Pang — Stanford University
- Letian He — Tsinghua University
- Chaoqi Liu — University of Illinois Urbana-Champaign
- Xiaoshen Han — Shanghai Jiao Tong University
- Shaurya Agrawal — Massachusetts Institute of Technology
- Susie Lu — Massachusetts Institute of Technology

Master's Students

- Haoyu Chen — Harvard University
- Feiyang Wu — Harvard University
- Alec Laprevotte — Harvard University

Ph.D. Students

- Yandong Ji — University of California San Diego

AWARDS AND HONORS

Outstanding Reviewer, Conference on Robot Learning (CoRL)	2026
Finalist - Best/Outstanding Paper Awards, Robotics: Science and Systems	2026
Conference Travel Award, Graduate College, University of Illinois Urbana-Champaign	2023
Finalist - Best Paper/Best Student Paper Awards, Conference on Robot Learning	2023
Bronze Tablet Award, University of Illinois Urbana-Champaign	2020
Excellent Graduate, Zhejiang University	2020
Student Speaker at Graduation, ZJU-UIUC Institute	2020
Provincial Government Scholarship, Zhejiang Province, China (Top 3%)	*2019*
Dean's List, University of Illinois Urbana-Champaign	2019
National Scholarship, China (Top 1.8%)	2018
Excellent Undergraduate Scholarship, Zhejiang University (Top 3%)	2018-2019
Academic Excellence Scholarship Award, ZJU-UIUC Institute	2017-2019
Top Ten Social Practice, Zhejiang University (Top 1.5%)	2017

SERVICE AND LEADERSHIP

- **IROS 2026 Workshop on Compositional and Modular Learning in the Era of Scaling in Robotics**
  Co-organizer · 2026
  Workshop Website

- **CoRL 2026 Workshop on Bimanual Dexterous Tool Use: From Contact-Rich Execution to Generalist Robot Intelligence**
  Co-organizer · 2026
  Workshop Website

- **CoRL 2026 Workshop on Compositional Generalization for Real-World Robot Learning**
  Co-organizer · 2026
  Workshop Website

- **ICRA 2025 Workshop on Learning Meets Model-Based Methods for Contact-Rich Manipulation** — Lead Organizer · May 2025
  Workshop Website
  Successfully Raised $1,400 in sponsorship from three companies and three IEEE RAS Technical Committees. The workshop attracted over 300 attendees.

- **Robotics Seminar @ Illinois** — Student Co-Organizer · August 2023–May 2025
  Seminar Series

- **Coordinated Science Laboratory Student Conference** — Reception Chair · 2024
  Conference Website

PROFESSIONAL SERVICE

- **Session Chair:**
  - IEEE International Conference on Robotics and Automation (ICRA), 2025 – Language-Guided Manipulation
  - CASE - IEEE Conference on Automation Science and Engineering, 2023
- **Reviewer:**
  - IEEE RAL - IEEE Robotics and Automation Letters
  - CoRL - Conference on Robot Learning
  - ICLR - International Conference on Learning Representations
  - ICRA - IEEE International Conference on Robotics and Automation
  - ITSC - IEEE International Conference on Intelligent Transportation Systems
  - CASE - IEEE Conference on Automation Science and Engineering

OUTREACH AND EXHIBITIONS

Sweet Gestures: Interactive Cupcake Sprinkling Robot 2024

- **Engineering Open House 2024** (Urbana-Champaign, IL)
  Designed and showcased robotic exhibit for 20,000+ attendees, engaging K-12 students from across Illinois
- **Society of Women Engineers WE24 Conference** (Chicago, IL)
  Demonstrated interactive robotics platform to 11,000+ conference participants

STEM Outreach Talk May 2025

Toward Human-Robot Alignment through Human Behavior Priors

Presented to Grade 5 students at St. Matthew Catholic School, fostering early interest in robotics and AI through accessible storytelling.

EXTRA-CURRICULAR SERVICE

Founding Team Leader, Robotic Team for RoboMaster Challenge
ZJU-UIUC Institute 10/2017–06/2018